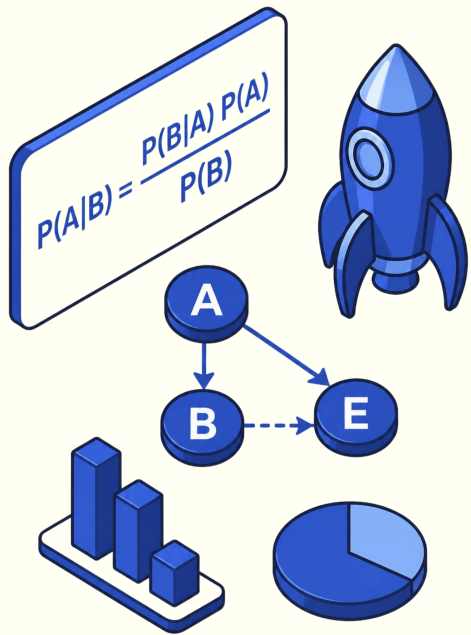




Naïve Bayes in Machine Learning

A Probabilistic Supervised Learning Algorithm for Classification

Blue-white AI-themed visual — probability graphics, Bayes formula accent, and clean vector motif for an academic classroom.

What is Naïve Bayes?

- Probabilistic supervised learning classifier based on Bayes' theorem
- "Naïve" = assumes feature independence given the class
- Outputs class posterior probabilities, not just labels
- Lightweight and fast for training and prediction



Spam Detection

Common real-world use — classifies emails using word frequencies.



Medical Diagnosis

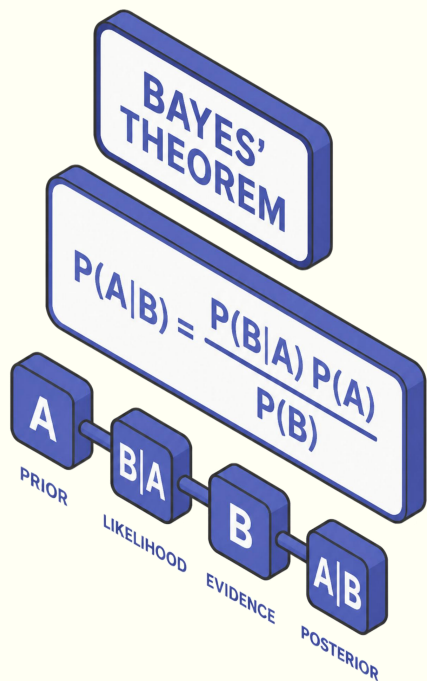
Predicts disease presence from symptoms (probabilistic support).



Sentiment & Document Classification

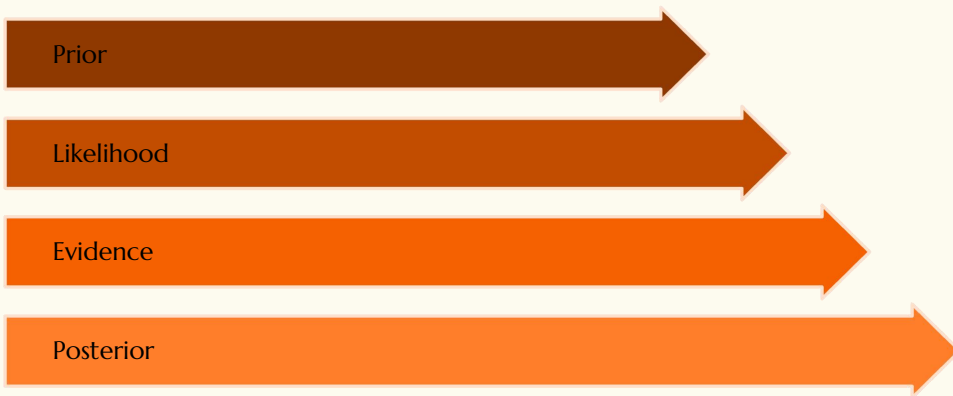
Fast baseline for NLP tasks with bag-of-words features.

Bayes' Theorem — Intuition & Formula



$$P(A|B) = P(B|A) \times P(A) / P(B)$$

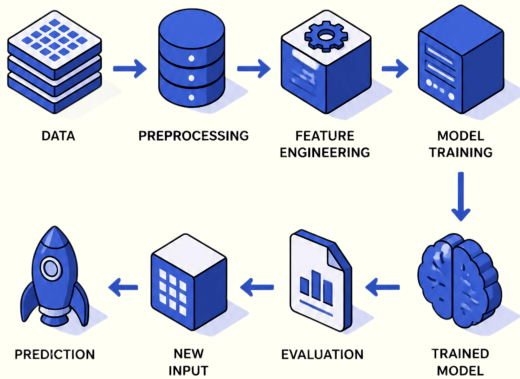
- **Prior** $P(A)$: belief before seeing data
- **Likelihood** $P(B|A)$: how likely data is under the class
- **Evidence** $P(B)$: overall probability of the data
- **Posterior** $P(A|B)$: updated belief after seeing data



Example: Updating disease probability after a positive test (use numeric example in lecture).

How Naïve Bayes Works — Step-by-Step

1. Collect labeled training data (features, class labels)
2. Estimate class priors $P(\text{Class})$
3. Estimate likelihoods $P(\text{feature}|\text{Class})$ for each feature
4. Compute posterior using Bayes' theorem for each class
5. Predict the class with highest posterior probability



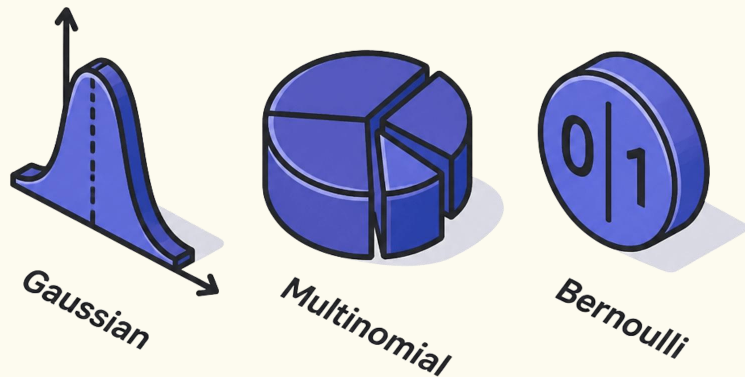
Training Data

Estimate Priors

Bayes Calculation

Prediction

Speaker note: Walk through a short numeric example (two classes, two features) to demonstrate calculations live.



Types of Naïve Bayes

Gaussian Naïve Bayes

For continuous numeric features; models likelihood with Gaussian (normal) distributions.

Multinomial Naïve Bayes

For count data (word frequencies); widely used in text classification.

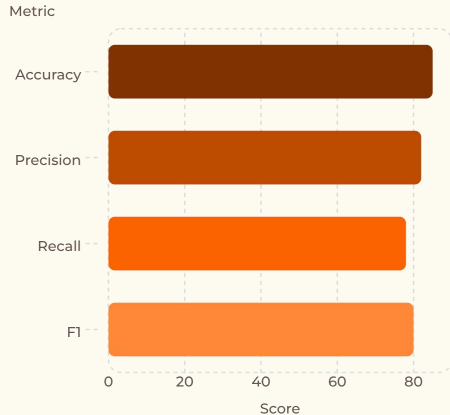
Bernoulli Naïve Bayes

For binary features (presence/absence); good for binary word occurrence.

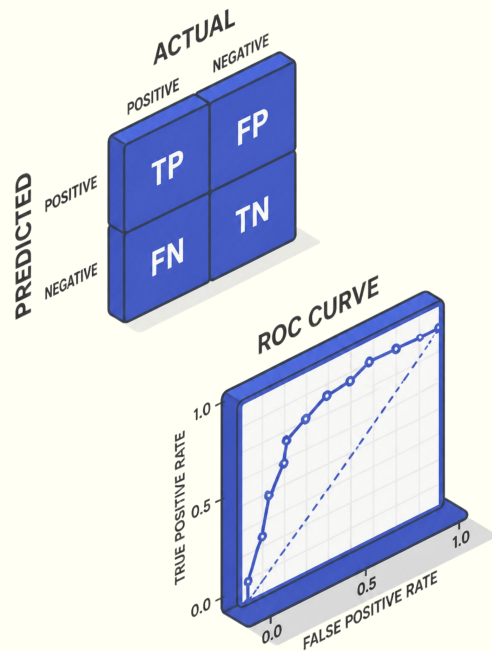
Tip: Choose variant based on data type and feature representation.

Model Evaluation Metrics

- Accuracy — overall correct predictions
- Precision — correctness among predicted positives
- Recall — coverage of actual positives
- F1 Score — harmonic mean of precision & recall
- Confusion Matrix — true/false positives and negatives
- ROC Curve & AUC — trade-off across thresholds



Cross-validation improves generalization estimates. Discuss threshold selection and class imbalance remedies (e.g., class priors, resampling).



Advantages & Limitations

Fast & Simple

Trains quickly; low memory

Works Well for Text

Effective with high-dimensional sparse data

Requires Less Data

Good baseline for many problems

Independence Assumption

May fail when features are correlated

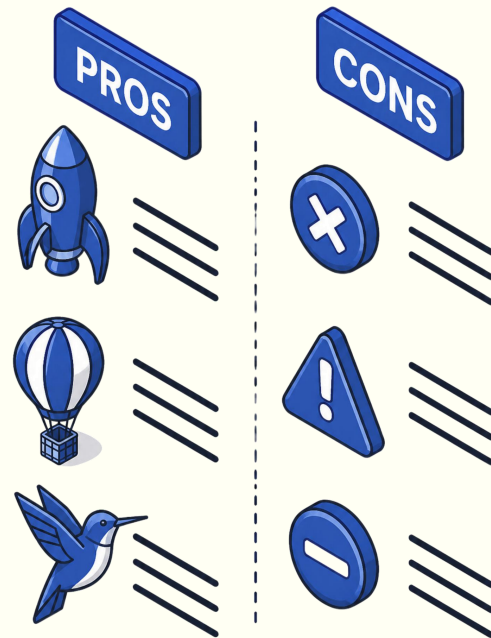
Zero-Frequency Problem

Features unseen in training → zero likelihood

Probability Estimates

Posterior probabilities can be poorly calibrated

Mitigation: Use Laplace (add-one) smoothing; feature selection; calibration methods (Platt scaling, isotonic regression).



Real-World Applications



Healthcare

Disease screening,
diagnostic
support



Finance

Fraud detection,
credit risk signals



Email Services

Spam filtering and
priority routing



NLP

Sentiment
analysis, language
detection



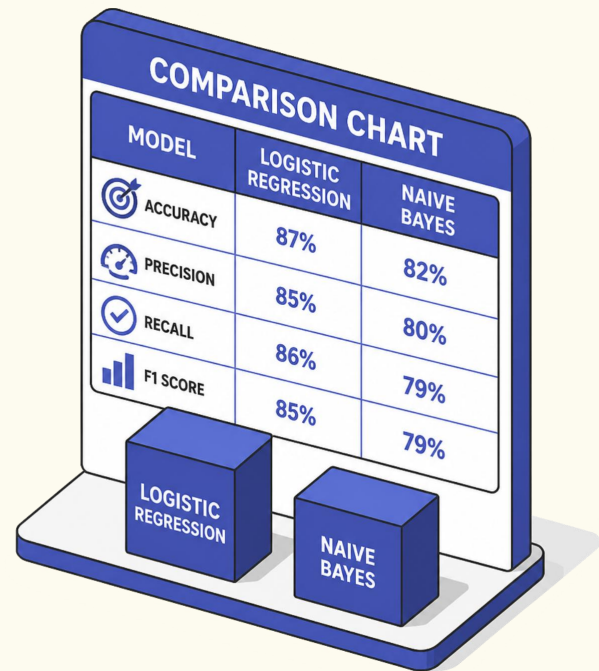
Cybersecurity

Malware and
intrusion
detection
heuristics

Speaker note: Emphasize how simplicity enables deployment at scale in industry systems.

Naïve Bayes vs Logistic Regression

Aspect	Naïve Bayes / Logistic Regression
Learning Type	Generative / Discriminative
Assumptions	Feature independence / Fewer parametric assumptions
Training Speed	Very fast / Moderate
Prob. Output	Posterior via Bayes (may be miscalibrated) / Well-calibrated with regularization
Best For	Text, high-dim sparse data / When feature interactions matter



Speaker note: Show an example where NB outperforms LR on small-sample text tasks, but LR wins when features interact and more labeled data is available.

Summary & Key Takeaways



- Naïve Bayes = fast, probabilistic supervised classifier based on Bayes' theorem
- Strong independence assumption simplifies computation — both strength and weakness
- Excellent baseline for text classification and high-dimensional problems
- Use Laplace smoothing and calibration when needed
- Compare with discriminative models (e.g., logistic regression) depending on data and task

"Naïve Bayes combines the power of probability and statistics to make fast, efficient, and surprisingly accurate predictions."

Thank you — Questions?

- 📝 Speaker notes: Wrap up with recommendations for experiments (compare variants, smoothing, feature engineering). Leave 3–5 minutes for Q&A.